

Trend-following Strategies for Crypto Investors

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August 2023

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ABSTRACT:

In the rapidly evolving landscape of financial markets, the emergence of cryptocurrencies as a distinct asset class has opened up new avenues for investors. However, investors need to establish a comprehensive investment strategy and implement risk-management measures before entering this space to mitigate the potential for market crashes. This paper develops and evaluates trend-following investment strategies in the context of cryptocurrencies. Secondly, it investigates the commonly held belief of a correlation between the movements of the Nasdaq 100 index and cryptocurrencies, which has significant implications for investment strategy development. We find that trend following strategies perform well over the researched period. Moreover, the effect of transaction costs is very substantial on portfolio performances. We also find no correlation between Nasdaq 100 index and our investigated cryptos, inconsistent with common perception in the market.

Key Takeaways:

- Trend-following strategies performed well over the analysed period
- Transaction costs significantly impact portfolio performances
- No correlation was found between the Nasdaq 100 index and the investigated cryptocurrencies, contrary to common market perceptions.

Keywords: Cryptocurrency, Trend-following strategy, Transaction cost, Risk management

JEL Classification: E42, E51, G11, G12, G21, G23

1. INTRODUCTION

The rise of cryptocurrencies, as a unique asset class, in the financial markets has opened up exciting opportunities for investors. Although cryptocurrencies are gaining popularity around the globe, they are highly volatile assets (Kim et al., 2021; Katsiampa, 2017). The price swings of cryptos are extremely higher than that in the money market. For example, in December 2017, Bitcoin prices rose by approximately 2700% to reach a record high of \$19,891, followed by the growth of other cryptocurrencies. Despite experiencing extreme volatility and substantial price swings, the largest cryptocurrencies have historically produced exponential returns (Koutmos, 2018; Katsiampa, 2019; Inci and Lagasse, 2019; Fintech Weekly, 2022). As a result, these assets have become appealing to investors who envision the possibility of profiting from their remarkable growth. However, it is essential for investors to establish a comprehensive investment strategy and implement risk-management measures before entering this space in order to mitigate the potential for market crashes. This paper develops and evaluates trend-following investment strategies in the context of cryptocurrencies. Secondly, it investigates the commonly held belief of a correlation between the movements of the Nasdaq 100 index and cryptocurrencies, a notion that has significant implications for investment strategy development. Addressing these issues is crucial as it not only contributes to the understanding of the dynamics of the cryptocurrency market but also provides practical insights for investors navigating this volatile yet potentially lucrative asset class.

Cryptocurrencies, notably Bitcoin (BTC) and Ether (ETH), have garnered significant attention in the past decade (Glenn and Reed, 2023). Since the inception of BTC, the crypto space has evolved from being a mere fintech concept (Nakamoto, 2008) to a distinct asset class. The introduction of Bitcoin futures contracts in December 2017 and the subsequent approval of Bitcoin futures ETFs by the US Securities and Exchange Commission (SEC) in late 2021 have brought Bitcoin closer to integration within the global financial system. Additionally, Bitcoin has been classified as a commodity under the Commodity Exchange Act, drawing similarities to gold in terms of its capacity as safe haven asset capacities (Selmi, Mensi, Hammoudeh, & Bouoiyour, 2018), as a medium of exchange (Kang and Lee, 2022), as well as its potential for hedging and diversification against traditional investments (Dyhrberg, 2016; Bedowska-Sójka and Kliber, 2022;

Enilov and Mishra, 2023). Next to Bitcoin is Ether - a native coin of Ethereum, a blockchain that hosts various other cryptocurrencies and begets a massive ecosystem of decentralized applications. Ether can be collected from investors in exchange for exercising smart contracts. The utility and value of Ether increased over time amid the worldwide popularity of the Ethereum platform.

Recently, Harvey et al. (2022) investigated 22 cryptocurrencies available at Coindesk from their first trading date to January 2022. They demonstrate that the annualised volatilities of typical cryptocurrencies available at Coinbase such as Ethereum, Litecoin, Solana etc. are significantly high. Most of them are greater than 100%, except for Bitcoin with 73.5%. Despite their high volatility, major cryptocurrencies have historically yielded exponential returns, attracting investors seeking significant growth. However, due to potential market crashes, it's crucial for investors to adopt robust investment strategies and risk-management measures. In this paper, we develop some trend-following strategies with zero and positive cash rates. In particular, we use the past 20, 65, 150, and 200-day price changes to create trading signals and define our portfolios. We show that trend-following strategies, especially those with a short lookback window, outperformed a constant investment in crypto, regardless of having zero or a positive cash rate. Additionally, we provide evidence of the impact of transaction costs on the performance of these strategies. Transaction costs erode part of the excess returns associated with our constructed portfolio because the high volatility of crypto assets, such as BTC and ETH, requires frequent portfolio rebalancing where the transaction costs occur.

We focus our analysis on Bitcoin and Ether as together they account for approximately 60% of the total crypto market (Encila, 2022). We also investigate whether the performance of other relatively small cryptocurrencies follows the same pattern as that of the two dominants. We use the S&P LargeCap-Ex. MegaCap index which constitutes digital assets with the largest market capitalisation, excluding BTC and ETH, and construct the same trend-following strategies as described above. We find similar results that trend-following strategies on this S&P index perform well over the researched period. Moreover, the effect of transaction costs is very significant on portfolio performances. The higher the transaction costs, the lower the excess returns.

Moreover, we examine whether it is possible to generate price signals and hence develop trad-

ing strategies for cryptocurrency from like asset classes. We evaluate the correlation between Nasdaq 100 index and our investigated cryptos. We choose Nasdaq 100 index for two reasons. First, given that crypto falls under the fintech category, Nasdaq 100 index seems most comparable to crypto since the index covers mostly tech-based companies. Second, average investors commonly believe that it is possible to use Nasdaq 100 index's movements to predict the price changes of crypto, specifically BTC, as they are highly correlated (Bonelli, 2020; Yaffe-Bellany, 2022). We find no correlation between Nasdaq 100 index and our investigated cryptos, inconsistent with common perception in the market.

Our research makes two contributions. First, we extend the literature on trend-following strategy. The last century has seen consistent and attractive returns from trend following in several markets (i.e., commodities, currencies, stock indices, bonds, and cryptocurrencies). For example, Lempérière et al. (2014), Hurst, Ooi & Pedersen (2017) observed excess returns based on trend-following strategies across different asset classes over the past two centuries. Szakmary et al. (2010) investigate the performance of trend following strategies in commodity futures and find that these pure trend-following strategies yield relatively higher mean returns and Sharpe ratios than momentum strategies. Rozario et al. (2020) and Harvey et al. (2022) showcase the potential returns in the cryptocurrency trend-following. We demonstrate that trend-following strategies performed well, even in the existence of transaction costs. Our findings show that due to the dynamic movements of cryptocurrencies, a short lookback window would produce better performance portfolios.

Second, our research also adds to the debate on the correlation between cryptos and other financial asset classes. Many recent studies (Conrad et al., 2018; Jiang et al., 2021; Lahiani and Jlassi, 2021; Dasman, 2021) explores the impact of cryptocurrency and the stock market in advanced and emerging economies. For example, Bouri et al. (2018) show that Bitcoin returns are closely associated with most of the other assets, especially commodities. Baumöhl (2019) finds a negative relationship between forex and cryptos, suggesting that investors can diversify their portfolios by investing in these assets alternatively. Other studies (such as Ghorbel and Jeribi, 2021) and news articles have also examined how cryptos relate to liken assets such as Nasdaq, S&P500. Coindesk illustrates that Bitcoin increased in tandem with the Nasdaq to S&P500 or NDX/SPX. Our research

investigates whether investors are able to create investment strategies for cryptos from Nasdaq 100 index. We find that Nasdaq 100 index is not a good predictor for crypto returns because there is no correlation between Nasdaq 100 index and our investigated cryptos.

The remainder of the paper proceeds as follows. Sections 2 and 3 describe the data and summary statistics. Sections 4, 5, and 6 show our analysis and Section 7 concludes the work.

2. DATA

We obtain historical prices of the Bitcoin, Ether, and LargeCap-Ex. MegaCap indexes from the S&P website which provides index series of digital asset performance covered by Lukka Prime. The bitcoin index sample covers 1,848 daily price observations from 01 January 2016 to 31 January 2023. The Bitcoin price started at USD57.77 and ended at USD3,079.23. The Ether index sample covers 1,581 observations from 10 January 2017 to 31 January 2023. The Ether price started at USD95.12 and ended at USD14,121.91. The LargeCap-Ex. MegaCap index sample covers 1,066 observations from 01 January 2019 to 31 January 2023. The index price started at USD25.37 and ended at USD48.65. The lookback periods applied for our analyses are 20, 65, 150, and 200 days. Details are provided in Section 4.

3. SUMMARY STATISTICS

Exhibit 10 illustrates the distribution of returns of the S&P Bitcoin, Ether, and LargeCap-Ex. MegaCap indexes over the sample period of 2016-2023. The top panel shows that Bitcoin's daily returns fall around plus and minus 20%. The same pattern applies to the S&P LargeCap-Ex. MegaCap index, which is reported in the bottom panel. For Ether, the daily returns fluctuate between plus and minus 40% approximately. Moreover, we find that these three instruments displayed non-normality. and Ether distributions are both right skewed with more clustering around their shoulders. Extreme tails happen more frequently than a Gaussian kernel-density estimator would

be able to capture. In contrast, the S&P LargeCap-Ex. MegaCap index exhibits a left-skewed distribution with deviations around the tails¹.

Next, we compute the monthly returns of the three indices. Exhibit 1 shows that Bitcoin, Ether, and the S&P LargeCap-Ex. MegaCap index share similar pattern over the sample years. Monthly returns of the three indexes fluctuate mostly in the same direction. For example, all three indices witnessed negative returns from April 2022 to December 2022, except for July and October 2022. This suggests that cryptocurrencies move much in unison in the market.

4. TREND FOLLOWING STRATEGY

4.1 Long only and momentum with a zero cash rate

Trend-following has existed and been profitable for decades. A trend-following investment involves going long markets with bullish movements and going short markets with bearish movements. Moskowitz, Ooi & Pedersen (2012) show that 12-month time series momentum provides a positive predictor of future return for each of the 58 examined asset instruments, i.e., equity index, currency, commodity, and bond futures. Prior studies (Szakmary et al., 2010; Lempérière et al. 2014; Hurst, Ooi & Pedersen, 2017; Rozario et al., 2020; and Harvey et al., 2022) find that trend-following strategy performed well throughout the past century across various markets.

We follow the literature to construct different trend strategies at the 20, 65, 150, and 200-day horizons and compare them with a buy-and-hold strategy. We compute the average price of instrument X over the past n days, where $X = \{\text{BTC, ETH, S\&P LargeCap-Ex. MegaCap index}\}$, $n = \{20, 65, 150, 200\}$:

$$\bar{X}_{t-1} = \frac{1}{n}(X_{t-1} + X_{t-2} + \dots + X_{t-n+1}) \quad (1)$$

We generate a rebalancing signal for an investment portfolio denoted as R . R is an indicator that equals 1 if the investor exits his current position and 0 otherwise. We then develop the

¹To save space, we do not report the plots of normal quantile and standardized normal probability and the results of Shapiro-Wilk W, Skewness and Kurtosis tests which indicate the essence of deviations and the rejection of normality.

following 20-day, 65-day, 150-day, and 200-day momentum-long-only strategies:

$$r_{t,momn-long} = \begin{cases} r_{t,X} & \text{If } X_{t-1} > \bar{X}_{t-1} \text{ and } R = 0; \text{ or } X_{t-1} \leq \bar{X}_{t-1} \text{ and } R = 1 \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

Where $r_{t,x}$ is the daily crypto return at day t . We report the basic performance statistics of the above-mentioned strategies in Exhibit 2.

The trend-following strategies outperform a constant investment in BTC, ETH, or the S&P LargeCap-Ex. MegaCap indexes in terms of performance characteristics such as annualised return, annualised volatility, Sharpe, Omega, and Sortino ratios, maximum drawdown and Ret/Max.drawdown over the period available to us. Moreover, the figures in Exhibit 3 suggest that the shorter lookback period gives us the most profitable strategy where all the performance ratios appeal to investors, i.e., higher return, lower volatility, and lower drawdown. For example, for all three indices, the 20-day and 65-day-momentum strategies are better performers than their corresponding counterparts. However, different from ETH and S&P LargeCap-Ex. MegaCap index, the 65-day momentum strategy, not the 20-day momentum strategy, produces the highest performance ratios for BTC.

4.2 Long only and momentum with a positive cash rate

In this section, we assume the cash rate ($r_{t,Cash}$) is the corresponding daily Treasury bill rate. We construct our strategies as follows:

$$r_{t,momn-long} = \begin{cases} r_{t,X} & \text{If } X_{t-1} > \bar{X}_{t-1} \text{ and } R = 0; \text{ or } X_{t-1} \leq \bar{X}_{t-1} \text{ and } R = 1 \\ r_{t,Cash} & \text{otherwise} \end{cases} \quad (3)$$

We show that having a positive cash rate does not lead to a significant change in the performance of our strategies. Similar to the previous section, we find that Investors are still able to earn excess returns by entering the coin space with a trend-following strategy. Moreover, using a trading signal over a short time window (65 days for BTC, 20 days for ETH and the S&P LargeCap-Ex MegaCap index) to restructure portfolios allows investors to acquire higher risk-adjusted returns.

4.3 Transaction cost-adjusted portfolios (with a positive cash rate)

The trading strategies analysed above do not incorporate trading costs. However, it can be a substantial part of the cost of managing a crypto portfolio. Previous studies, such as (Korajczyk, 2004), (Lesmond, Schill, & Zhou, 2004), address the importance of transaction costs on the profitability of momentum-based trading strategies in stock portfolios. They suggest that abnormal profits generated from these investments can disappear depending on the magnitude of trading costs. There are varying levels of trading costs across different cryptocurrency platforms or exchanges. Therefore, it is crucial to analyse the effect of these transaction costs on the portfolios' expected returns.

In this section, we analyse the performance of our long-only and momentum strategies after adjusting for transaction costs. Transaction costs are normally modest and vary from 0.05% to up to 4% across different cryptocurrency platforms or exchanges. For example, Binance.com charges 0.1% for an investor's position, whereas he needs to pay a 0.5% transaction fee to trade on Bitstamp. We employ transaction costs of 0.1%, 0.25%, and 0.5% of the trading volume and report the results in Exhibit 4. The results for strategies with zero and positive cash rates are relatively similar. Therefore, to save space, we only report the latter in Exhibit 4.

We find that transaction costs erase some of the profits from investing in BTC and ETH, but not all of them. Taking portfolios in BTC as an example, the Sharpe ratio of Mom20-long without transaction cost is 1.68, while that of the same strategy with 0.5% transaction cost is 1.25. Likewise, investors appear to achieve abnormal returns when trading the S&P LargeCap-Ex. MegaCap index, even after accounting for trading commissions. Notably, all four strategies demonstrate higher Sharpe ratios compared to the buy-and-hold portfolio, indicating the potential for enhanced risk-adjusted performance through active trading in this index. Lastly, we suggest that the high volatility of BTC and ETH requires frequent portfolio rebalancing relative to the S&P LargeCap-Ex.MegaCap index, such that the effect of transaction costs on price behavior is larger. Therefore, the existence of transaction costs reduces the profitability of these investments.

We also estimate the cumulative returns (CR) from these strategies for the different coins. We find that for all these coins, the short window momentum long-only strategy produces much higher cumulative returns than the buy-and-hold strategy. This is shown in Exhibit 6A (Appendix). We find that a dollar invested in BTC in January 2016 has a CR of 3,913.85% (11,437.75%) by January 2023 following the 20 (65)-day momentum strategy while the buy and hold strategy produces only 5,257.98% CR. The corresponding numbers for ETH and S&P LargeCap-Ex. MegaCap index following the 20-day momentum strategy are 54,757.47% and 406.45% respectively.

5. RISK OVERLAY STRATEGY (WITH A POSITIVE CASH RATE)

5.1 Drawdown-driven strategy

In this section, we construct investment strategies based on the daily drawdown of the price of X. We compare the daily drawdown at day $t-1$ with the corresponding volatility calculated by rolling n -day returns. The strategies are formulated as follows:

$$r_{t,DDn} = \begin{cases} r_{t,X} & \text{If } DD_{t-1} < 50\%Vol_{t-1} \\ r_{t,Cash} & \text{otherwise} \end{cases} \quad (4)$$

where DD_{t-1} is a daily drawdown of X at day $t-1$; Vol_{t-1} is a daily volatility at day $t-1$ using daily returns in the window of 20, 65, 150, and 200 days.

We also investigate the effect of transaction costs on the performance of these strategies. Again, we apply 0.1%, 0.25%, and 0.5% of the trading volume and report the results in Exhibit 6. We show that all four investments generated significantly lower returns compared to a buy-and-hold investment. However, their Sharpe ratios (maximum drawdowns) were relatively higher (lower) than the corresponding counterparts. This suggests that these drawdown-driven strategies are the

potential to construct investment portfolios in crypto.

5.2 Constant annualised volatility strategy

Next, we analyse a volatility-targeted strategy. We calculate the rolling n - day volatility, $n = \{20, 65, 150, 200\}$. We annualise volatility by multiplying with the square root of 261, equivalent to the number of days in a year. The portfolios are constructed as follows:

$$r_{t,CVn} = w_{t-1,X}r_{t,X} + w_{t-1,Cash}r_{t,Cash}$$

where $w_{t,X} = \text{Max} \left(\frac{0.3}{\text{Rollingndayvolatility}_{t-1} * \sqrt{261}}, 100\% \right)$; $w_{t,Cash} = 1 - w_{t,X}$; 0.3 is the *target annualised volatility*; $r_{t,Cash}$ is the corresponding daily treasury bill rate at day t .

Exhibit 7 report the performance ratios of the 30% target volatility portfolios on BTC, ETH, and the S&P LargeCap-Ex. MegaCap index. Similar to the prior result, we find that these strategies performed quite nicely to create a higher Sharpe ratio and lower drawdown compared to a buy-and-hold investment. Moreover, interestingly, when we increased the target volatility, the Sharpe ratios also increased.

5.3 Long only, momentum and constant annualised volatility strategy

In this section, we overlay the momentum strategy discussed in section 4 on the volatility-targeted strategy. We use equation (1) to compute the average price of X instruments over the past n days. We then develop the following 20-day, 65-day, 150-day, and 200-day momentum with a target volatility of 30% p.a. strategies:

$$r_{t,CBn} = w_{t-1,X}r_{t,X} + w_{t-1,Cash}r_{t,Cash}$$

where

$$w_{t,X} = \begin{cases} \text{Max} \left(\frac{0.3}{\text{Rolling } n \text{ day volatility}_{t-1} * \sqrt{261}}, 100\% \right), & \text{If } X_{t-1} > \bar{X}_{t-1} \text{ and } R = 0; \text{ or } X_{t-1} \leq \bar{X}_{t-1} \text{ and } R = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$w_{t,Cash} = 1 - w_{t,X};$$

We report the performance statistics of the above-constructed portfolios in Exhibit 8. Exposing to a risk of 30% and looking back at a shorter period brought investors to less hazardous investments with higher Sharpe ratios. Once again, the results support the dynamic behaviour of crypto prices, suggesting that investors better look at a short time horizon, e.g., 65 days for BTC and 20 days for ETH and S&P, to create their investment strategies.

6. CORRELATION BETWEEN CRYPTO AND NASDAQ 100 INDEX

The crypto market posted massive fluctuations over its short history. Average investors may worry about the probability of dramatic losses when a crypto bubble bursts. Therefore, they may seek alternatives in the traditional financial markets to invest or diversify their investments. Given that cryptocurrency, with its backbone technology and the blockchain, falls under the fintech category, we believe that Nasdaq 100 index is one of the most comparable assets to crypto. This is because the index covers more than 100 of the largest public companies, mostly technology-based, such as Apple, Amazon, and Tesla, to name just a few. In 2018, those blue-chip companies comprise 56.14% of the index, followed by consumer services with 24.9%. In this section, we conjecture whether Nasdaq has any correlations with our investigated crypto instruments. Roth, S. (2022, 9) shows that between January 2020 and June 2020, the correlation between Nasdaq and Bitcoin increased from -0.1 to 0.801. Sui Chung, CEO of CF Benchmarks, explained: *"The reason correlations move higher is that the same type of investor begins to invest in two different assets. So, if you take that as market maturation, correlations will start to look very similar."*

We perform a Granger causality test on the assets over the sample period. To be comparable,

we exclude public holidays during the investigated time window when the stock market is closed, but not the crypto market. We present the results in Exhibit 9.

Inconsistent with recent literature which finds a stable and strong contemporaneous correlation between BTC and Nasdaq 100 index (Bonelli, 2020; Yaffe-Bellany, 2022), we find that Nasdaq's returns do not granger cause crypto returns (i.e., BTC return, ETH return, and S&P LargeCap-Ex MegaCap index return). Specifically, in Exhibit 9, our coefficients are statistically insignificant, suggesting the acceptance of the null hypothesis. This indicates that lagged values of Nasdaq return do not statistically explain the movement of BTC return (ETH return and S&P LargeCap-Ex MegaCap index return) and vice versa.

7. CONCLUSION

This paper contributes to the literature on trend-following strategies in cryptocurrency markets by constructing and evaluating different portfolios based on the S&P Bitcoin, Ether, and LargeCap-Ex. MegaCap indexes, with zero and positive cash rates. Our main findings are as follows. First, we show that both individual and institutional investors can benefit from applying a momentum (risk overlay) strategy to invest in crypto assets, as it allows them to capture the upside potential and avoid the downside risk of these volatile instruments. Second, we find that a shorter lookback horizon of 20 days is more effective than a longer one of 60 days, as it adapts better to the dynamic movements and volatility of crypto prices. Third, we account for the impact of transaction costs (up to 0.5%) on the performance of our strategies and demonstrate that they significantly reduce the abnormal returns of the portfolios, especially for those with higher rebalancing frequency. This implies that investors should carefully consider the trade-off between transaction costs and portfolio rebalancing when implementing trend-following strategies in crypto markets.

Furthermore, we examine the correlation between Nasdaq 100 and crypto assets over our sample period. Contrary to common perception and recent literature, we do not find any significant correlation between the Nasdaq index and any of the crypto instruments we investigate, suggesting that the return patterns of crypto assets cannot be explained by the performance of the Nasdaq

index and vice versa.

In conclusion, investing in crypto assets exposes investors to various sources of risk. To mitigate these risks, investors should adopt a comprehensive investment strategy that incorporates trend-following rules and considers the effect of transaction costs if applicable. Our paper provides empirical evidence and insights for investors who are interested in participating in the crypto space.

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Exhibit 1: Monthly returns

Panel A: Monthly BTC returns%													
%	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	YTD
2016	-13.21	16.29	-4.61	9.17	14.94	28.06	-1.56	-12.84	5.41	16.02	5.92	28.86	121.57
2017	-0.38	24.67	-9.00	21.69	73.70	8.97	13.89	66.08	-11.82	52.37	51.15	50.03	1400.27
2018	-30.44	6.18	-36.27	37.31	-19.01	-22.01	31.77	-9.13	-5.28	-5.47	-37.23	-7.02	-74.46
2019	-6.92	10.99	7.29	28.32	61.47	46.30	-19.04	-4.09	-13.71	11.70	-16.26	-7.78	94.21
2020	30.22	-7.35	-24.86	36.40	6.33	-2.79	24.25	3.36	-8.77	26.69	43.18	50.32	308.31
2021	18.64	33.85	26.15	-2.33	-35.38	-5.75	14.27	19.05	-7.96	43.72	-8.13	-20.19	57.22
2022	-16.11	8.96	8.75	-15.95	-17.55	-40.20	26.51	-15.41	-3.66	4.55	-16.04	-3.15	-63.89
2023	39.45												39.45
Panel B: Monthly ETH returns%													
%	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	YTD
2017	0.76	48.79	203.14	39.13	237.05	26.19	-31.06	94.81	-24.33	4.88	36.36	74.31	6713.36
2018	52.30	-21.57	-56.58	79.62	-15.23	-28.71	6.11	-34.81	-21.66	-10.60	-42.34	18.50	-81.37
2019	-19.75	27.35	3.51	15.87	56.53	19.43	-30.68	-21.66	5.53	2.99	-15.92	-17.23	-5.86
2020	41.13	24.08	-40.11	57.78	3.92	2.37	53.24	26.92	-18.79	8.08	57.82	23.14	483.97
2021	81.10	7.08	31.26	46.83	-5.75	-14.61	5.85	43.83	-12.41	47.25	5.54	-21.62	388.52
2022	-26.29	5.55	15.82	-14.97	-30.80	-47.23	69.14	-8.65	-15.11	16.89	-17.16	-7.60	-67.16
2023	32.60												32.60
Panel C: Monthly S&P LargeCap-Ex. MegaCap index returns%													
%	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	YTD
2019	-16.71	16.75	17.15	14.50	46.96	-2.30	-24.77	-22.86	-11.68	16.10	-17.90	-15.08	-22.31
2020	51.45	-9.18	-30.43	24.76	-3.78	-3.84	43.73	13.97	-19.04	-1.25	65.24	-20.12	90.92
2021	45.04	102.88	17.24	40.33	-31.00	-20.15	-0.34	70.33	-12.52	25.25	2.05	-14.67	331.97
2022	-28.91	0.48	16.09	-21.77	-41.54	-32.40	34.56	-11.08	-2.69	5.19	-18.12	-18.09	-78.94
2023	42.13												42.13

Note: The exhibit shows the monthly returns of the S&P Bitcoin, Ether and LargeCap-Ex. MegaCap Indexes. Data are obtained from S&P, covering from 01 January 2016 to 31 January 2023 for Bitcoin, from 10 January 2017 to 31 January 2023 for Ether, and from 31 January 2023 for the S&P LargeCap-Ex MegaCap index.

Exhibit 2: Long only and momentum with a zero cash rate

Panel A: BTC					
Performace Statistics	<i>Buy and Hold</i>	<i>Mom20- long</i>	<i>Mom65- long</i>	<i>Mom150- long</i>	<i>Mom200- long</i>
Annualised return (%)	75.47	91.25	105.20	84.59	81.01
Annualised volatility (%)	71.25	53.79	55.71	57.10	58.84
Annualised Sharpe Ratio (Rf=0%)	1.06	1.70	1.89	1.48	1.38
Omega Ratio	1.23	1.44	1.46	1.38	1.34
Sortino Ratio	1.73	2.36	2.52	2.11	2.03
Maximum drawdown (%)	82.97	72.54	57.88	62.56	66.68
Ret/Max. drawdown	0.91	1.26	1.82	1.35	1.21
Panel B: ETH					
Performace Statistics	<i>Buy and Hold</i>	<i>Mom20- long</i>	<i>Mom65- long</i>	<i>Mom150- long</i>	<i>Mom200- long</i>
Annualised return (%)	128.87	213.66	161.14	168.58	130.58
Annualised volatility (%)	103.23	81.08	85.13	86.38	86.81
Annualised Sharpe Ratio (Rf=0%)	1.25	2.64	1.89	1.95	1.50
Omega Ratio	1.27	1.59	1.46	1.46	1.40
Sortino Ratio	2.12	3.26	2.62	2.65	2.33
Maximum drawdown (%)	93.45	52.89	61.44	78.73	89.76
Ret/Max. drawdown	1.38	4.04	2.62	2.14	1.45
Panel C: S&P LargeCap-Ex MegaCap index					
Performace Statistics	<i>Buy and Hold</i>	<i>Mom20- long</i>	<i>Mom65- long</i>	<i>Mom150- long</i>	<i>Mom200- long</i>
Annualised return (%)	18.48	66.37	39.84	18.92	25.13
Annualised volatility (%)	83.13	58.83	61.05	61.72	59.42
Annualised Sharpe Ratio (Rf=0%)	0.22	1.13	0.65	0.31	0.42
Omega Ratio	1.12	1.32	1.24	1.16	1.18
Sortino Ratio	0.88	1.77	1.24	0.83	0.96
Maximum drawdown (%)	84.34	60.98	62.78	59.16	61.82
Ret/Max.drawdown	0.22	1.09	0.63	0.32	0.41

Note: The exhibit shows the performance statistics of various long-only and momentum strategies on BTC, ETH and the S&P LargeCap-Ex. MegaCap index versus a corresponding buy-and-hold investment. Data are obtained from S&P, covering from 01 January 2016 to 31 January 2023 for BTC, from 10 January 2017 to 31 January 2023 for ETH, and from 01 January 2019 to 31 January 2023 for the S&P LargeCap-Ex MegaCap index.

Exhibit 3: Long only and momentum with a positive cash rate

Panel A: BTC						
Performance Statistics	<i>Buy and Hold</i>	<i>Mom20-long</i>	<i>Mom65-long</i>	<i>Mom150-long</i>	<i>Mom200-long</i>	
Annualised return (%)	75.47	92.02	106.17	85.52	81.87	
Annualised volatility (%)	71.25	53.78	55.71	57.10	58.84	
Annualised Sharpe Ratio (Rf=0.83%)	1.04	1.68	1.88	1.47	1.37	
Panel B: ETH						
Performance Statistics	<i>Buy and Hold</i>	<i>Mom20-long</i>	<i>Mom65-long</i>	<i>Mom150-long</i>	<i>Mom200-long</i>	
Annualised return (%)	128.87	215.19	162.44	170.18	131.99	
Annualised volatility (%)	103.23	81.08	85.13	86.37	86.81	
Annualised Sharpe Ratio (Rf=0.93%)	1.23	2.62	1.88	1.94	1.50	
Panel C: S&P LargeCap-Ex MegaCap index						
Performance Statistics	<i>Buy and Hold</i>	<i>Mom20-long</i>	<i>Mom65-long</i>	<i>Mom150-long</i>	<i>Mom200-long</i>	
Annualised return (%)	18.48	67.16	40.59	19.70	26.02	
Annualised volatility (%)	83.13	58.83	61.05	61.72	59.42	
Annualised Sharpe Ratio (Rf=0.87%)	0.21	1.12	0.65	0.30	0.42	

Note: The exhibit shows the performance statistics of various long-only and momentum strategies on BTC, ETH and the S&P LargeCap-Ex. MegaCap index versus a corresponding buy-and-hold investment where a positive cash rate is incorporated. Data are obtained from S&P, covering from 01 January 2016 to 31 January 2023 for BTC, from 10 January 2017 to 31 January 2023 for ETH, and from 01 January 2019 to 31 January 2023 for the S&P LargeCap-Ex MegaCap index.

Exhibit 4: Transaction cost-adjusted portfolio

Panel A: BTC												
Transaction	Mom20-long			Mom65-long			Mom150-long			Mom200-long		
Cost	0.1%	0.25%	0.5%	0.1%	0.25%	0.5%	0.1%	0.25%	0.5%	0.1%	0.25%	0.5%
Annualised return (%)	87.07	79.87	68.45	104.03	100.81	95.54	84.56	83.12	80.73	80.83	79.27	76.68
Annualised volatility (%)	53.79	53.80	53.85	55.70	55.70	55.71	57.10	57.09	57.09	58.84	58.85	58.86
Annualised Sharpe Ratio (Rf=0.83%)	1.59	1.46	1.25	1.84	1.78	1.69	1.45	1.43	1.39	1.35	1.32	1.28
Panel B: ETH												
Transaction	Mom20-long			Mom65-long			Mom150-long			Mom200-long		
Cost	0.1%	0.25%	0.5%	0.1%	0.25%	0.5%	0.1%	0.25%	0.5%	0.1%	0.25%	0.5%
Annualised return (%)	208.54	198.82	183.27	159.00	153.85	145.47	169.26	167.86	165.54	130.68	128.69	125.40
Annualised volatility (%)	81.09	81.10	81.14	85.11	85.09	85.07	86.38	86.39	86.40	86.82	86.83	86.85
Annualised Sharpe Ratio (Rf=0.93%)	2.54	2.42	2.23	1.84	1.78	1.68	1.93	1.91	1.89	1.48	1.46	1.42
Panel C: S&P LargeCap-Ex. MegaCap index												
Transaction	Mom20-long			Mom65-long			Mom150-long			Mom200-long		
Cost	0.1%	0.25%	0.5%	0.1%	0.25%	0.5%	0.1%	0.25%	0.5%	0.1%	0.25%	0.5%
Annualised return (%)	63.32	57.71	48.76	39.35	37.46	34.37	19.10	18.18	16.67	25.12	23.74	21.48
Annualised volatility (%)	58.83	58.84	58.88	61.05	61.07	61.10	61.72	61.73	61.74	59.42	59.42	59.43
Annualised Sharpe Ratio (Rf=0.87%)	1.05	0.96	0.81	0.62	0.59	0.54	0.29	0.28	0.25	0.40	0.38	0.34

Note: The exhibit shows the performance statistics of cost-based portfolios on BTC, ETH and the S&P LargeCap-Ex. MegaCap index. Data are obtained from S&P, covering from 01 January 2016 to 31 January 2023 for BTC, from 10 January 2017 to 31 January 2023 for ETH, and from 01 January 2019 to 31 January 2023 for the S&P LargeCap-Ex MegaCap index.

Exhibit 5: Monthly Returns of the Best Performer (with 0.5% transaction costs)

Panel A: Monthly BTC returns%													
%	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	YTD
2016	-10.24	3.56	-9.83	9.17	14.94	28.06	-1.56	-13.72	0.02	14.30	5.92	28.86	78.50
2017	-0.38	24.67	-10.75	13.77	73.70	8.97	19.64	66.08	-11.82	52.37	51.15	50.03	1345.16
2018	-20.99	0.09	0.10	2.87	-6.42	0.11	-6.53	-10.33	-12.94	0.14	-2.05	0.14	-45.322
2019	0.15	-3.72	7.29	28.32	61.47	46.30	-19.04	-3.98	-4.73	0.11	0.09	0.09	132.96
2020	13.11	-7.35	-8.28	-0.49	6.33	-2.79	24.25	3.36	-18.75	19.08	43.18	50.32	164.39
2021	18.64	33.85	26.15	-4.61	-15.45	0.00	5.36	15.06	-13.34	29.49	-8.06	-15.38	71.01
2022	0.01	0.02	1.94	-15.06	0.06	0.09	0.13	0.17	-2.43	-0.90	-21.65	0.26	-33.92
2023	18.11												18.11
Panel B: Monthly ETH returns%													
%	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	YTD
2017	0.76	48.79	203.14	19.46	237.05	4.56	0.06	75.83	-29.38	0.85	11.57	74.31	4562.45
2018	52.34	-15.28	-3.11	32.13	1.25	0.11	-18.84	0.13	-11.51	-11.16	-17.60	19.90	5.71
2019	-6.58	11.62	-4.56	14.76	41.00	15.35	-12.30	-8.78	-20.43	-4.31	-3.56	0.09	9.23
2020	26.11	28.38	0.02	26.80	-8.31	4.47	35.32	11.88	-26.59	-1.63	57.82	20.96	310.47
2021	81.10	18.03	-11.65	46.83	-0.47	0.00	-3.83	43.83	-19.09	28.26	2.02	-11.37	258.21
2022	0.01	-16.78	1.28	-9.91	0.06	0.09	16.53	-9.54	-8.07	2.95	-20.51	-8.39	-44.74
2023	26.05												26.05
Panel C: Monthly S&P LargeCap-Ex. MegaCap index returns%													
%	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	YTD
2019	-9.08	8.66	14.99	15.64	17.24	-1.94	0.13	-0.71	-27.64	-9.01	-8.77	0.09	-9.73
2020	32.50	8.12	0.02	14.10	-12.47	0.25	28.92	8.31	-11.60	-6.24	57.35	-22.27	103.07
2021	22.09	102.88	-6.87	33.13	1.18	0.00	3.45	70.33	-12.45	5.77	2.24	-9.66	368.32
2022	-6.13	-19.83	12.65	-16.66	0.06	0.09	-0.95	-8.93	-7.85	0.37	-20.42	-4.12	-54.96
2023	31.62												31.62

Note: This exhibit reports monthly returns of the 65-day momentum and long only portfolio for BTC, and the 20-day momentum and long only portfolio for ETH and the S&P LargeCap-Ex MegaCap index. Data are obtained from S&P, covering from 01 January 2016 to 31 January 2023 for BTC, from 10 January 2017 to 31 January 2023 for ETH, and from 01 January 2019 to 31 January 2023 for the S&P LargeCap-Ex MegaCap index.

Exhibit 6: Drawdown-driven strategy

Panel A: BTC																
Transaction	DD20				DD65				DD150				DD200			
Cost	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%
Annualised return (%)	39.47	37.74	35.19	31.04	38.98	37.28	34.74	30.60	36.52	35.00	32.73	29.02	35.35	33.81	31.50	27.73
Annualised volatility (%)	28.53	28.53	28.54	28.57	28.61	28.60	28.60	28.62	28.39	28.39	28.41	28.46	28.35	28.35	28.37	28.42
Annualised Sharpe Ratio (Rf=0.83%)	1.34	1.28	1.19	1.05	1.32	1.26	1.18	1.03	1.25	1.19	1.11	0.98	1.21	1.15	1.07	0.94
Panel B: ETH																
Transaction	DD20				DD65				DD150				DD200			
Cost	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%
Annualised return (%)	40.42	38.90	36.64	32.95	54.95	53.29	50.80	46.73	46.20	44.72	42.51	38.90	42.61	41.12	38.90	35.27
Annualised volatility (%)	42.66	42.65	42.63	42.61	42.99	42.97	42.94	42.90	42.84	42.83	42.82	42.80	42.60	42.59	42.59	42.59
Annualised Sharpe Ratio (Rf=0.83%)	0.92	0.88	0.83	0.74	1.25	1.21	1.15	1.06	1.05	1.01	0.96	0.88	0.97	0.94	0.88	0.80
Panel C: S&P LargeCap-Ex. MegaCap index																
Transaction	DD20				DD65				DD150				DD200			
Cost	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%
Annualised return (%)	16.65	15.29	13.27	9.98	21.12	19.67	17.48	13.93	22.16	20.76	18.64	15.19	16.60	15.32	13.38	10.22
Annualised volatility (%)	29.54	29.51	29.46	29.40	29.97	29.92	29.86	29.77	30.12	30.07	30.00	29.92	29.29	29.26	29.23	29.20
Annualised Sharpe Ratio (Rf=0.83%)	0.53	0.48	0.42	0.31	0.67	0.62	0.55	0.44	0.70	0.66	0.59	0.47	0.53	0.49	0.42	0.32

Note: The exhibit shows the performance statistics of drawdown-driven strategy on BTC, ETH and the S&P LargeCap-Ex. MegaCap index. We compute daily volatility by overlapping 20, 65, 150 or 200-day returns. Data are obtained from S&P, covering from 01 January 2016 to 31 January 2023 for BTC, from 10 January 2017 to 31 January 2023 for ETH, and from 01 January 2019 to 31 January 2023 for the S&P LargeCap-Ex MegaCap index.

Exhibit 7: Constant volatility strategy

Panel A: BTC																
Transaction	CV20				CV65				CV150				CV200			
Cost	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%
Annualised return (%)	48.74	47.93	46.73	44.73	44.99	44.73	44.36	43.72	43.64	43.53	43.37	43.10	44.15	44.07	43.95	43.76
Annualised volatility (%)	35.39	35.40	35.41	35.42	32.95	32.95	32.95	32.95	31.47	31.47	31.47	31.47	31.05	31.05	31.05	31.05
Annualised Sharpe Ratio (Rf=0.83%)	1.34	1.32	1.29	1.23	1.33	1.32	1.31	1.29	1.35	1.35	1.34	1.33	1.38	1.38	1.38	1.37
Panel B: ETH																
Transaction	CV20				CV65				CV150				CV200			
Cost	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%
Annualised return (%)	48.71	48.04	47.04	45.40	41.07	40.91	40.66	40.26	42.18	42.10	41.99	41.81	40.95	40.90	40.82	40.69
Annualised volatility (%)	35.54	35.54	35.54	35.54	32.21	32.21	32.21	32.21	31.66	31.66	31.66	31.66	30.70	30.70	30.70	30.70
Annualised Sharpe Ratio (Rf=0.83%)	1.33	1.31	1.29	1.24	1.23	1.23	1.22	1.21	1.29	1.29	1.29	1.28	1.29	1.29	1.29	1.28
Panel C: S&P LargeCap-Ex. MegaCap index																
Transaction	CV20				CV65				CV150				CV200			
Cost	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%
Annualised return (%)	30.10	29.46	28.50	26.92	14.84	14.69	14.46	14.09	16.08	16.02	15.92	15.76	15.11	15.06	14.99	14.87
Annualised volatility (%)	35.43	35.43	35.42	35.41	31.57	31.57	31.57	31.57	29.96	29.96	29.96	29.96	29.15	29.15	29.15	29.15
Annualised Sharpe Ratio (Rf=0.83%)	0.82	0.80	0.77	0.73	0.44	0.43	0.43	0.42	0.50	0.50	0.50	0.49	0.48	0.48	0.48	0.48

Note: The exhibit shows the performance statistics of constant volatility strategy on BTC, ETH and the S&P LargeCap-Ex. MegaCap index. We set our target volatility for our portfolio at 30%. We compute daily volatility for cryptos by overlapping 20, 65, 150 or 200-day returns. Data are obtained from S&P, covering from 01 January 2016 to 31 January 2023 for BTC, from 10 January 2017 to 31 January 2023 for ETH, and from 01 January 2019 to 31 January 2023 for the S&P LargeCap-Ex MegaCap index.

Exhibit 8: Long only, momentum and constant annualised volatility strategy

Panel A: BTC																
Transaction	CB20				CB65				CB150				CB200			
Cost	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%
Annualised return (%)	52.33	49.82	46.12	40.15	53.00	52.07	50.68	48.39	46.07	45.71	45.16	44.26	44.60	44.23	43.66	42.72
Annualised volatility (%)	27.62	27.64	27.66	27.72	26.37	26.38	26.38	26.40	25.48	25.48	25.48	25.48	25.99	25.99	25.99	26.00
Annualised Sharpe Ratio (Rf=0.83%)	1.85	1.76	1.62	1.41	1.96	1.93	1.87	1.79	1.76	1.75	1.73	1.69	1.67	1.66	1.63	1.60
Panel B: ETH																
Transaction	CB20				CB65				CB150				CB200			
Cost	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%
Annualised return (%)	61.76	60.10	57.66	53.66	44.96	44.27	43.24	41.54	45.96	45.75	45.43	44.91	37.80	37.52	37.11	36.42
Annualised volatility (%)	27.83	27.83	27.83	27.84	25.76	25.76	25.76	25.77	25.47	25.48	25.48	25.48	24.62	24.62	24.62	24.64
Annualised Sharpe Ratio (Rf=0.83%)	2.17	2.11	2.02	1.88	1.69	1.67	1.63	1.56	1.75	1.74	1.73	1.71	1.48	1.47	1.46	1.43
Panel C: S&P LargeCap-Ex. MegaCap index																
Transaction	CB20				CB65				CB150				CB200			
Cost	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%
Annualised return (%)	37.04	35.38	32.93	28.93	16.58	16.06	15.28	13.99	14.36	14.12	13.76	13.17	14.67	14.38	13.95	13.24
Annualised volatility (%)	27.10	27.10	27.10	27.10	23.82	23.83	23.84	23.86	22.09	22.09	22.09	22.10	20.46	20.46	20.46	20.47
Annualised Sharpe Ratio (Rf=0.83%)	1.32	1.26	1.17	1.03	0.65	0.63	0.60	0.55	0.61	0.59	0.58	0.55	0.67	0.66	0.63	0.60

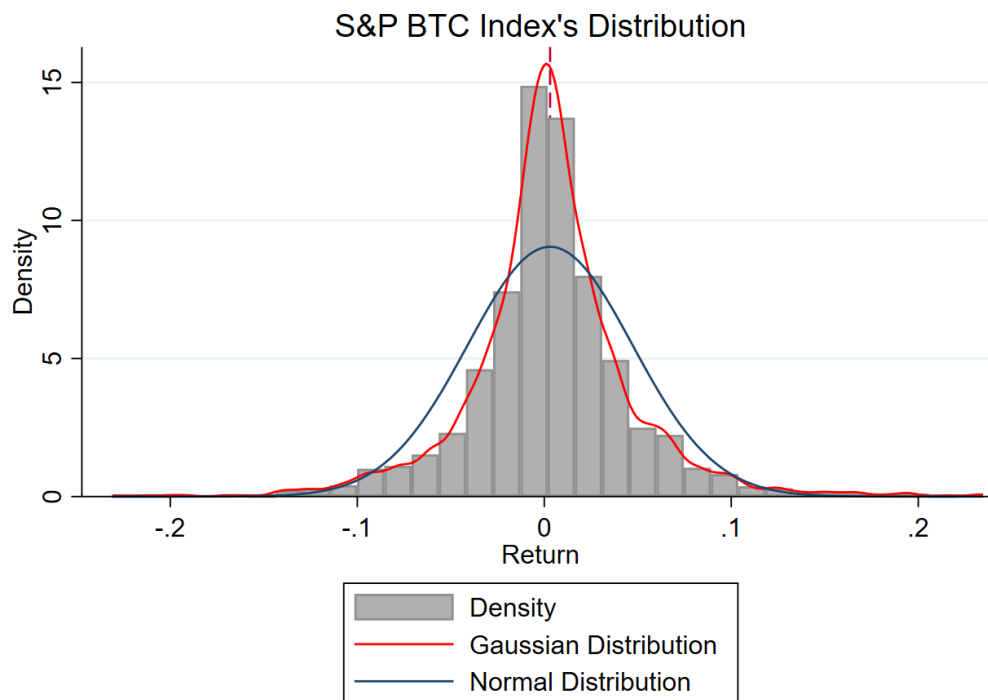
Note: The exhibit shows the performance statistics of long only, momentum and constant annualised volatility strategies on BTC, ETH and the S&P LargeCap-Ex. MegaCap index. We compute daily volatility by overlapping 20, 65, 150 or 200-day returns. Data are obtained from S&P, covering from 01 January 2016 to 31 January 2023 for BTC, from 10 January 2017 to 31 January 2023 for ETH, and from 01 January 2019 to 31 January 2023 for the S&P LargeCap-Ex MegaCap index.

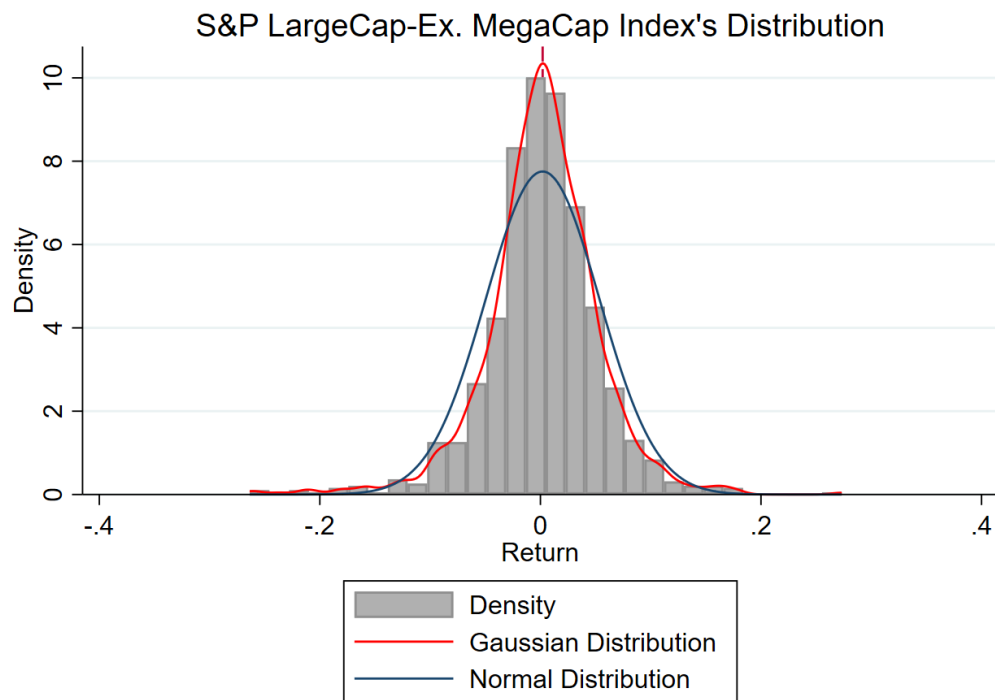
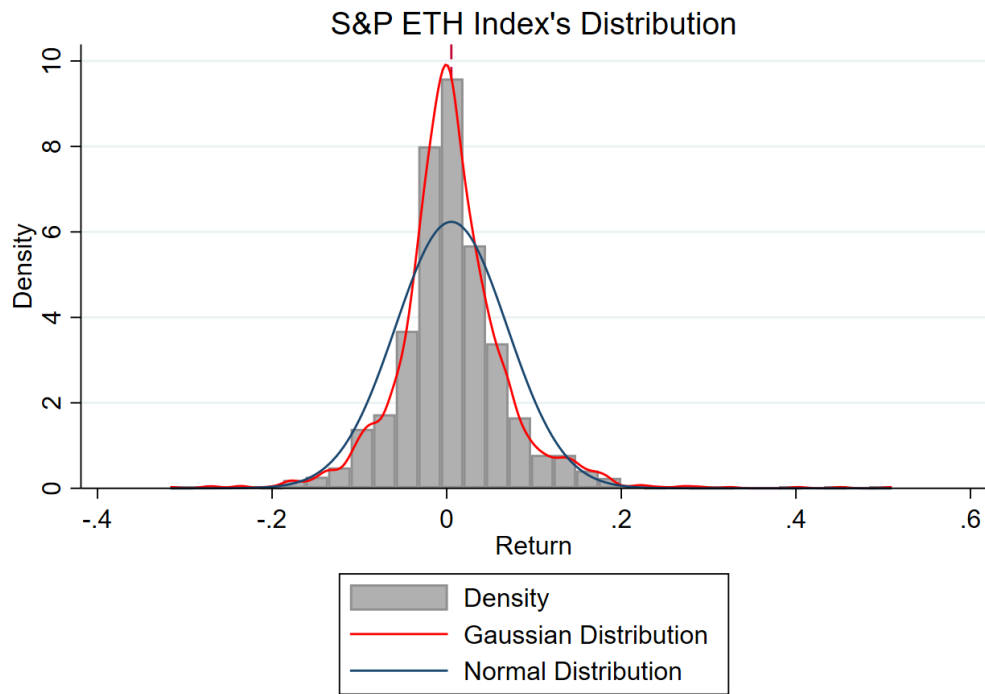
Exhibit 9: Granger causality Wald test

Panel A: Granger causality Wald test between BTC Index and Nasdaq			
Equation	Excluded	Chi-square	Prob>Chi-square
BTC return	Nasdaq return	0.0763	0.782
Nasdaq return	BTC return	0.0095	0.922
Panel B: Granger causality Wald test between ETH Index and Nasdaq			
Equation	Excluded	Chi-square	Prob>Chi-square
ETH return	Nasdaq return	0.2076	0.649
Nasdaq return	ETH return	1.8019	0.179
Panel C: Granger causality Wald test between S&P LargeCap-Ex. MegaCap Index and Nasdaq			
Equation	Excluded	Chi-square	Prob>Chi-square
Index return	Nasdaq return	0.1213	0.728
Nasdaq return	Index return	0.1300	0.718

Note: The exhibit shows the results of the Granger causality test between the Nasdaq 100 index and the three investigated crypto assets. Panels A reports the results between Nasdaq and BTC. Panel B shows the correlation between Nasdaq and ETH. Panel C shows the correlation between Nasdaq and the S&P LargeCap-Ex. MegaCap index. Data are obtained from S&P, covering from 01 January 2016 to 31 January 2023 for BTC, from 10 January 2017 to 31 January 2023 for ETH, and from 01 January 2019 to 31 January 2023 for the S&P LargeCap-Ex MegaCap index.

Exhibit 10: Histogram of the S&P Bitcoin, Ether and LargeCap-Ex. MegaCap Indexes





A. Appendix

Exhibit 1A: Long only and momentum with a positive cash rate: Other performance ratios

Panel A: BTC						
Performace Statistics	<i>Buy and Hold</i>	<i>Mom20-long</i>	<i>Mom65-long</i>	<i>Mom150-long</i>	<i>Mom200-long</i>	
Omega Ratio	1.23	1.44	1.46	1.38	1.34	
Sortino Ratio	1.73	2.38	2.53	2.12	2.04	
Maximum drawdown (%)	82.97	72.31	57.30	61.96	66.18	
Ret/Max. drawdown	0.91	1.27	1.85	1.38	1.24	
Panel B: ETH						
Performace Statistics	<i>Buy and Hold</i>	<i>Mom20-long</i>	<i>Mom65-long</i>	<i>Mom150-long</i>	<i>Mom200-long</i>	
Omega Ratio	1.27	1.59	1.46	1.47	1.40	
Sortino Ratio	2.12	3.27	2.63	2.66	2.34	
Maximum drawdown (%)	93.45	52.54	61.35	78.30	89.55	
Ret/Max. drawdown	1.38	4.10	2.65	2.17	1.47	
Panel C: S&P LargeCap-Ex MegaCap index						
Performace Statistics	<i>Buy and Hold</i>	<i>Mom20-long</i>	<i>Mom65-long</i>	<i>Mom150-long</i>	<i>Mom200-long</i>	
Omega Ratio	1.12	1.33	1.24	1.16	1.19	
Sortino Ratio	0.88	1.78	1.26	0.85	0.98	
Maximum drawdown (%)	84.34	60.64	62.50	59.16	61.82	
Ret/Max. drawdown	0.22	1.11	0.65	0.33	0.42	

Exhibit 2A: Transaction cost-adjusted portfolio

Panel A: BTC												
Transaction	Mom20-long			Mom65-long			Mom150-long			Mom200-long		
Cost	0.1%	0.25%	0.5%	0.1%	0.25%	0.5%	0.1%	0.25%	0.5%	0.1%	0.25%	0.5%
Omega Ratio	1.42	1.39	1.35	1.46	1.45	1.43	1.38	1.37	1.36	1.34	1.34	1.33
Sortino Ratio	2.29	2.17	1.97	2.50	2.45	2.37	2.11	2.09	2.05	2.02	2.00	1.96
Maximum drawdown (%)	73.23	74.57	77.29	57.88	58.75	60.18	62.18	62.52	63.08	66.44	66.84	67.50
Ret/Max. drawdown	1.19	1.07	0.89	1.80	1.72	1.59	1.36	1.33	1.28	1.22	1.19	1.14
Panel B: ETH												
Transaction	Mom20-long			Mom65-long			Mom150-long			Mom200-long		
Cost	0.1%	0.25%	0.5%	0.1%	0.25%	0.5%	0.1%	0.25%	0.5%	0.1%	0.25%	0.5%
Omega Ratio	1.58	1.56	1.53	1.46	1.45	1.43	1.46	1.46	1.46	1.40	1.39	1.39
Sortino Ratio	3.22	3.14	3.02	2.61	2.56	2.49	2.66	2.64	2.63	2.33	2.31	2.28
Maximum drawdown (%)	53.10	53.94	55.31	61.84	62.58	63.78	78.55	78.93	79.56	89.78	90.11	90.64
Ret/Max. drawdown	3.93	3.69	3.31	2.57	2.46	2.28	2.15	2.13	2.08	1.46	1.43	1.38
Panel C: S&P LargeCap-Ex. MegaCap index												
Transaction	Mom20-long			Mom65-long			Mom150-long			Mom200-long		
Cost	0.1%	0.25%	0.5%	0.1%	0.25%	0.5%	0.1%	0.25%	0.5%	0.1%	0.25%	0.5%
Omega Ratio	1.31	1.29	1.26	1.24	1.23	1.22	1.16	1.15	1.15	1.18	1.18	1.17
Sortino Ratio	1.72	1.62	1.47	1.24	1.20	1.15	0.84	0.82	0.79	0.96	0.93	0.89
Maximum drawdown (%)	61.88	63.67	66.47	63.04	63.86	65.20	59.24	59.36	60.52	61.97	62.20	62.58
Ret/Max. drawdown	1.02	0.91	0.73	0.62	0.59	0.53	0.32	0.31	0.28	0.41	0.38	0.34

Exhibit 3A: Drawdown-driven strategy: Other performance ratios

Panel A: BTC																
Transaction	DD20				DD65				DD150				DD200			
Cost	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%
Omega Ratio	2.08	2.02	1.93	1.79	2.04	1.98	1.90	1.77	1.98	1.93	1.86	1.74	1.96	1.90	1.83	1.71
Sortino Ratio	2.55	2.46	2.31	2.07	2.51	2.42	2.27	2.04	2.38	2.30	2.16	1.95	2.32	2.23	2.10	1.88
Maximum drawdown (%)	27.97	29.04	30.62	33.21	28.53	29.46	30.82	33.07	31.23	32.12	33.43	35.60	30.54	31.44	32.77	34.95
Ret/Max. drawdown	1.41	1.30	1.15	0.93	1.37	1.27	1.13	0.93	1.17	1.09	0.98	0.82	1.16	1.08	0.96	0.79
Panel B: ETH																
Transaction	DD20				DD65				DD150				DD200			
Cost	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%
Omega Ratio	1.83	1.80	1.75	1.68	2.11	2.07	2.01	1.92	1.93	1.90	1.85	1.77	1.88	1.85	1.80	1.72
Sortino Ratio	1.91	1.86	1.78	1.64	2.44	2.38	2.30	2.16	2.12	2.07	1.99	1.86	2.00	1.95	1.87	1.73
Maximum drawdown (%)	49.58	51.07	53.22	56.62	38.94	39.18	39.74	40.94	43.60	43.93	45.17	48.38	43.60	43.93	45.17	48.38
Ret/Max. drawdown	0.82	0.76	0.69	0.58	1.41	1.36	1.28	1.14	1.06	1.02	0.94	0.80	0.98	0.94	0.86	0.73
Panel C: S&P LargeCap-Ex. MegaCap index																
Transaction	DD20				DD65				DD150				DD200			
Cost	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%
Omega Ratio	1.51	1.47	1.41	1.32	1.59	1.55	1.49	1.40	1.62	1.58	1.52	1.43	1.49	1.46	1.40	1.32
Sortino Ratio	1.04	0.98	0.88	0.72	1.25	1.18	1.08	0.91	1.29	1.23	1.13	0.97	1.04	0.97	0.88	0.73
Maximum drawdown (%)	26.43	27.60	29.32	32.33	25.07	26.23	27.98	31.04	24.67	25.98	27.95	31.36	32.01	33.20	34.98	37.85
Ret/Max. drawdown	0.63	0.55	0.45	0.31	0.84	0.75	0.62	0.45	0.90	0.80	0.67	0.48	0.52	0.46	0.38	0.27

Exhibit 4A: Constant volatility strategy: Other performance ratios

Panel A: BTC

Transaction	CV20				CV65				CV150				CV200			
Cost	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%
Omega Ratio	1.28	1.27	1.27	1.26	1.27	1.27	1.27	1.27	1.27	1.27	1.27	1.27	1.28	1.28	1.28	1.28
Sortino Ratio	1.98	1.96	1.92	1.86	1.97	1.96	1.94	1.92	1.98	1.98	1.97	1.96	2.04	2.04	2.03	2.02
Maximum drawdown (%)	53.56	53.87	54.33	55.09	50.46	50.54	50.66	50.87	46.13	46.16	46.21	46.29	44.75	44.77	44.81	44.87
Ret/Max. drawdown	0.91	0.89	0.86	0.81	0.89	0.89	0.88	0.86	0.95	0.94	0.94	0.93	0.99	0.98	0.98	0.98

Panel B: ETH

Transaction	CV20				CV65				CV150				CV200			
Cost	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%
Omega Ratio	1.27	1.27	1.26	1.26	1.25	1.25	1.25	1.25	1.26	1.26	1.26	1.26	1.26	1.26	1.26	1.26
Sortino Ratio	2.04	2.02	1.99	1.94	1.95	1.95	1.94	1.92	2.07	2.07	2.06	2.06	2.05	2.05	2.04	2.04
Maximum drawdown (%)	63.84	63.99	64.20	64.56	52.30	52.34	52.39	52.49	50.32	50.33	50.36	50.41	47.82	47.84	47.86	47.89
Ret/Max. drawdown	0.76	0.75	0.73	0.70	0.79	0.78	0.78	0.77	0.84	0.84	0.83	0.83	0.86	0.85	0.85	0.85

Panel C: S&P LargeCap-Ex. MegaCap index

Transaction	CV20				CV65				CV150				CV200			
Cost	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%
Omega Ratio	1.18	1.18	1.17	1.17	1.11	1.11	1.11	1.11	1.12	1.12	1.12	1.12	1.12	1.12	1.12	1.12
Sortino Ratio	1.37	1.35	1.32	1.26	0.84	0.84	0.83	0.81	0.92	0.91	0.91	0.90	0.88	0.87	0.87	0.87
Maximum drawdown (%)	51.04	51.32	51.74	52.43	51.07	51.15	51.27	51.47	45.70	45.73	45.78	45.86	44.14	44.16	44.19	44.24
Ret/Max. drawdown	0.59	0.57	0.55	0.51	0.29	0.29	0.28	0.27	0.35	0.35	0.35	0.34	0.34	0.34	0.34	0.34

Exhibit 5A: Long only, momentum and constant annualized volatility strategy: Other performance ratios

Panel A: BTC																
Transaction	CB20				CB65				CB150				CB200			
Cost	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%
Omega Ratio	1.53	1.51	1.47	1.41	1.54	1.53	1.51	1.49	1.46	1.46	1.46	1.45	1.42	1.42	1.41	1.40
Sortino Ratio	2.75	2.64	2.48	2.22	2.87	2.83	2.76	2.66	2.56	2.54	2.52	2.48	2.46	2.45	2.42	2.38
Maximum drawdown (%)	52.06	52.95	54.41	57.77	29.15	29.71	30.54	31.91	25.40	25.79	26.36	27.43	26.14	26.68	27.48	28.79
Ret/Max. drawdown	1.01	0.94	0.85	0.70	1.82	1.75	1.66	1.52	1.81	1.77	1.71	1.61	1.71	1.66	1.59	1.48
Panel B: ETH																
Transaction	CB20				CB65				CB150				CB200			
Cost	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%
Omega Ratio	1.63	1.61	1.59	1.55	1.47	1.46	1.45	1.43	1.50	1.49	1.49	1.48	1.42	1.41	1.41	1.40
Sortino Ratio	3.34	3.27	3.17	2.99	2.68	2.64	2.59	2.51	2.84	2.83	2.82	2.79	2.44	2.42	2.40	2.36
Maximum drawdown (%)	27.62	27.99	28.53	29.42	26.84	27.38	28.17	29.48	29.70	29.91	30.24	30.78	39.21	39.65	40.29	41.36
Ret/Max. drawdown	2.24	2.15	2.02	1.82	1.67	1.62	1.53	1.41	1.55	1.53	1.50	1.46	0.96	0.95	0.92	0.88
Panel C: S&P LargeCap-Ex. MegaCap index																
Transaction	CB20				CB65				CB150				CB200			
Cost	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%	0%	0.1%	0.25%	0.5%
Omega Ratio	1.41	1.39	1.36	1.32	1.22	1.21	1.20	1.19	1.20	1.19	1.19	1.18	1.21	1.21	1.20	1.19
Sortino Ratio	2.11	2.03	1.91	1.72	1.12	1.09	1.05	0.97	1.05	1.04	1.02	0.98	1.11	1.10	1.07	1.02
Maximum drawdown (%)	30.74	31.72	33.17	35.52	36.22	36.69	37.39	38.55	23.45	23.81	24.33	25.20	23.59	23.88	24.30	24.99
Ret/Max. drawdown	1.20	1.12	0.99	0.81	0.46	0.44	0.41	0.36	0.61	0.59	0.57	0.52	0.62	0.60	0.57	0.53

Exhibit 6A: Cumulative returns of long-only and momentum portfolios (with cash rate and 0.5% transaction cost)

